Band-1 Hadamard Order 3188 — Miyamoto (H_796 Paley-II) + Paley-I Cross-Check; Gap 2052 Still OPEN

H_796/C_798/H_3188 PASS (Miyamoto + Paley-I); 3188 in SPK via Paley I q=3187; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3076_Miyamoto`; `2026-09-15_Band1_Hadamard_3044_Miyamoto`; `2026-09-15_Band1_Hadamard_2836_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3188_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 796 and 3188 only** (verified $\max|HH^\top - nI| = 0$), via **two independent paths** for order 3188 (Miyamoto primary; Paley type I cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3188 **enters SPK** via Paley I ($q = 3187$); it is **not** a Miyamoto-only bonus (contrast prior 3076/3044/2836/2692/2564). Honest SPK audit: Paley-II **FAIL** ($q = 1593$ not prime); Kronecker **FAIL** ($3188 = 4 \cdot 797$, 797 not Hadamard-admissible); Paley-I **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3076 on box (incl. Miyamoto-only bonuses 2836/3044/3076). This increment constructs the **next preferred** Miyamoto-eligible target after $q = 769$: **order 3188** ($3188 = 4 \cdot 797$, 797 prime $\equiv 1 \pmod 4$). Seed H_{796} via Paley type II ($q = 397$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 796$). Symmetric Paley conference C_{798} over prime 797. $H_{3188} = \text{Miyamoto}(H_{796}, C_{798})$. Independently, H_{3188} also verifies via **Paley type I** with $q = 3187$ (explains SPK membership). Unit tests **26/26 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3188; Miyamoto; Paley type II; Paley type I; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{796} constructed and verified (Paley type II, $q = 397$). 2. C_{798} constructed and verified (Paley conference, $q = 797$): $CC^\top = 797I$, zero diagonal. 3. H_{3188} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type I ($q = 3187$). **Not** the Hadamard conjecture. 4. SPK: $3188 \in$ SPK (via Paley I). Paley-II and Kronecker paths for 3188 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3188 = 4 \cdot 797$
$q = 797$	prime, $797 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{796}
Needed conference	order $798 = q + 1$
H_{796} path (used)	Paley type II, $q = 397$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 796$
Conference path	Symmetric Paley over prime 797
SPK path for 3188	Paley type I, $q = 3187$ prime $\equiv 3 \pmod 4$; order $q + 1 = 3188$
Paley-II for 3188	**REJECTED**: $q' = 1593$ not prime
Kronecker for 3188	**REJECTED**: only 4×797; $797 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{796}	Paley type II, $q = 397$	0	0.3016	**PASS**
C_{798}	Paley conference, $q = 797$	0	0.5376	**PASS**
H_{3188}	Miyamoto(H_{796}, C_{798})	0	12.9818	**PASS**
H_{3188} P-I	Paley type I, $q = 3187$	0	17.8508	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-I cross-check) $\approx$ 32.9754 s. Compressed ± 1 matrices saved as `H796_pm1.npz`, `C798_pm1.npz`, `H3188_pm1.npz`, `H3188_paleyI_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior ships $\cup$ verified H_{796} $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3188 enters SPK **because** Paley I applies ($q = 3187$).

Quantity	Value
Known count	885
3188 in SPK	**True** (Paley I $q = 3187$)
796 in SPK	True (Paley II $q = 397$)
3076 in SPK	False (prior Miyamoto bonus)
3044 in SPK	False (prior Miyamoto bonus)
2836 in SPK	False (prior Miyamoto bonus)
2804 in SPK	True (prior Paley I)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284, 2292, 2300…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (26/26)

Test	Result
`H4_sylvester`	PASS

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H20_paley_I_control	PASS
paley_II_q5_H12	PASS
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q11_H12	PASS
q397_is_prime	PASS
q397_equiv_1_mod_4	PASS
H796_paley_II_order_formula	PASS
q797_is_prime	PASS
q797_equiv_1_mod_4	PASS
n3188_equals_4_times_797	PASS
H796_order_formula	PASS
C798_order_formula	PASS
paley_I_H3188_path_eligible	PASS
paley_II_H3188_path_rejected	PASS
kronecker_H3188_path_rejected	PASS
q3187_is_prime_equiv_3_mod_4	PASS
paley_I_order_3188_formula	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_3188_arith	PASS
conference_798_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
3188_in_spk_via_paley_I_arith	PASS
paley_II_seed_pattern_control	PASS

7. Next optional targets

Preferred next Miyamoto-eligible primes after $q = 797$ (build notes from on-box triage):

- $q = 809 \to n = 3236$ (need H_{808}); notes: Kronecker(404x2)
- $q = 821 \to n = 3284$ (need H_{820}); notes: PaleyII(q=409)
- $q = 829 \to n = 3316$ (need H_{828}); notes: PaleyI(q=827)
- $q = 853 \to n = 3412$ (need H_{852}); seed path TBD
- $q = 857 \to n = 3428$ (need H_{856}); notes: Kronecker(428x2)
- $q = 877 \to n = 3508$ (need H_{876}); seed path TBD

Also optional: method-attack SPK gap **2052** (GS/SDS / non-Paley conference); Track B privacy / Object 1 meter await. GATE-04 remains **LOCKED**.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 3188 is **in SPK** (Paley I); not a Miyamoto-only bonus.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Artifacts

Lab: `/workspace/experiments/band1/hadamard_3188_miyamoto/`
Verifier: `verify_hadamard_3188.py`
Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
Matrices: `H796_pm1.npz`, `C798_pm1.npz`, `H3188_pm1.npz`, `H3188_paleyI_pm1.npz`
Shelf: `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_3188_Miyamoto.pdf`